

WHATSNEXT VISION MOTORS

SALESFORCE CRM PROJECT

Project Documentation

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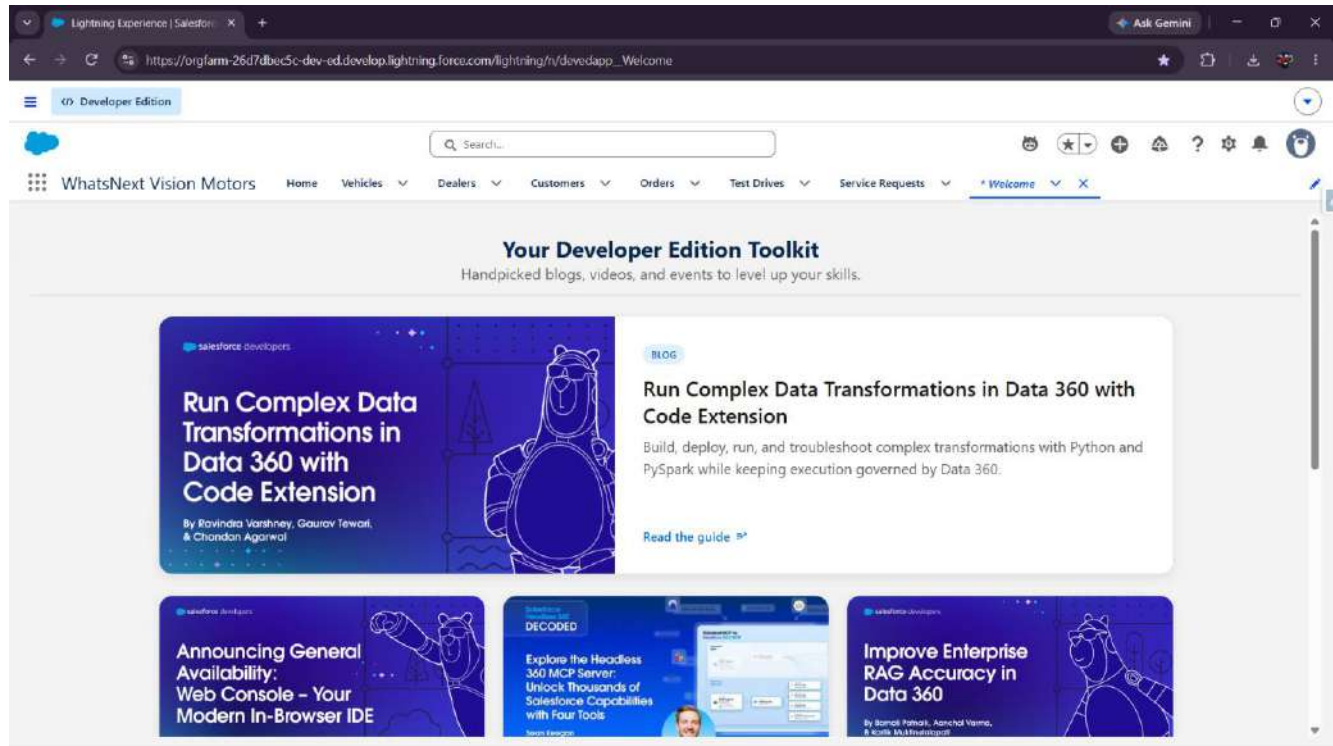
Branch & Year: CSB - 4th Year

Salesforce Trailhead Profile:

<https://www.salesforce.com/trailblazer/dyjyzvcoc58m0ilgdf>

1. Project Overview

WhatsNext Vision Motors is a Salesforce-based Customer Relationship Management (CRM) application designed to manage vehicle inventory, dealer information, customers, vehicle orders, test drives, and service requests. The project demonstrates Salesforce data modeling, Lightning application development, Flow automation, Apex programming, Batch Apex, and Scheduled Apex.



2. Problem Statement

Vehicle dealerships need an organized system to manage vehicle stock, customer information, dealer details, orders, test drives, and service requests. Manual processing can result in incorrect orders, delayed follow-ups, and difficulty identifying the appropriate dealer. This project addresses these requirements using Salesforce CRM and automation.

3. Objectives

- Create a centralized Salesforce CRM application for vehicle dealership operations.
- Maintain vehicle details and real-time stock availability.
- Store dealer information including location details.
- Manage customer information and contact details.
- Create and process vehicle orders.
- Prevent customers from ordering vehicles that are out of stock.
- Automatically assign the nearest dealer based on customer and dealer location.
- Send automated reminders for scheduled test drives.
- Automatically update service request status.

- Periodically synchronize vehicle availability using Batch Apex and Scheduled Apex.
- Automatically process pending orders when vehicle stock becomes available.
- Send scheduled notifications for stock replenishment and pending order processing.

4. Project Requirements

4.1 Functional Requirements

- Vehicle Management – Create and maintain vehicle model, brand, price, stock quantity, and availability.
- Dealer Management – Maintain dealer name and location information including city, state, street, latitude, and longitude.
- Customer Management – Store customer name, address, phone, email, and location information.
- Order Management – Create orders with customer, vehicle, dealer, quantity, date, status, and total amount.
- Stock Validation – Block an order when the selected vehicle has zero stock.
- Nearest Dealer Assignment – Automatically select the nearest dealer using customer and dealer latitude/longitude.
- Test Drive Management – Schedule test drives and maintain customer, vehicle, dealer, date/time, status, and reminder information.
- Test Drive Reminder – Send an automated email reminder one hour before a scheduled test drive.
- Service Request Management – Create service requests and automatically move the status to In Progress.
- Vehicle Stock Batch Processing – Periodically update vehicle Availability based on Stock Quantity.
- Order Status Processing – Change pending orders to Confirmed when stock is available.
- Scheduled Notifications – Send notifications for out-of-stock vehicles and pending orders.

4.2 Non-Functional Requirements

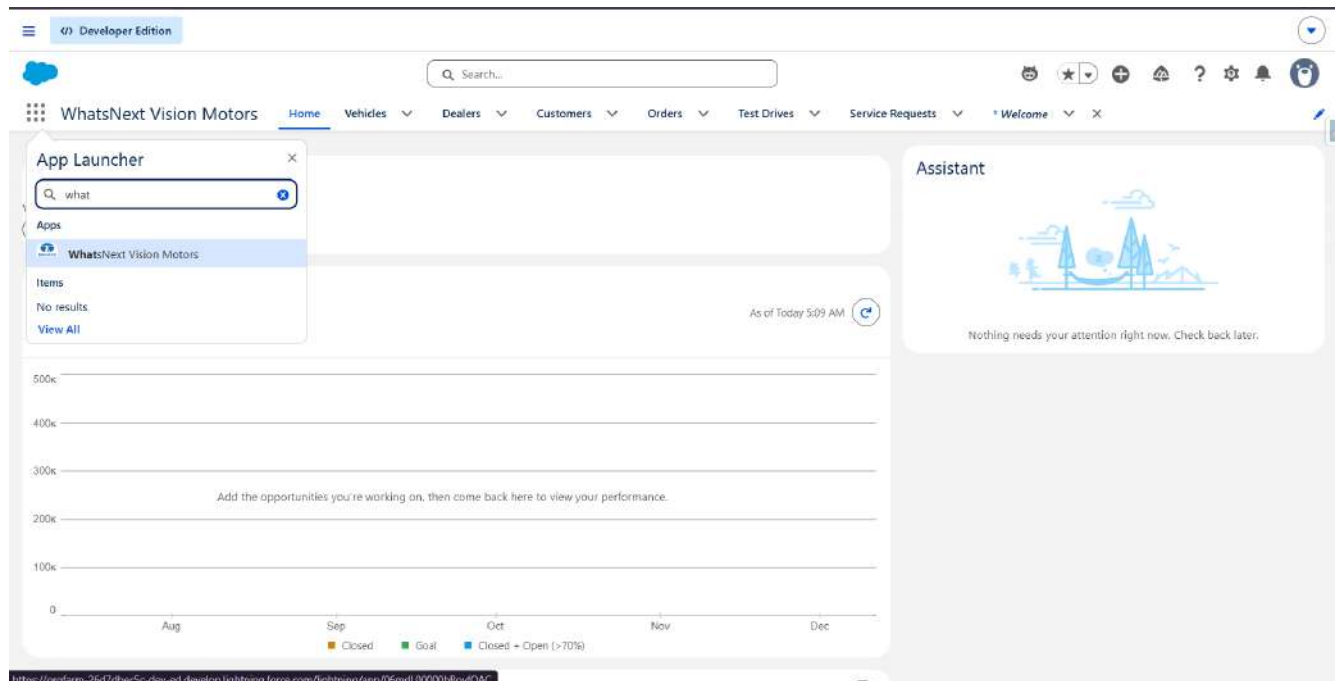
- The application should be easy to use through Salesforce Lightning.
- Data should be organized using appropriate Salesforce objects and relationships.
- Automation should reduce manual processing.
- Apex logic should be reusable through a trigger handler.
- Batch processing should support multiple records efficiently.
- The system should provide clear validation messages for invalid orders.

5. Technologies Used

- Salesforce CRM
- Salesforce Lightning App Builder
- Salesforce Objects, Fields and Relationships
- Record-Triggered Flows
- Apex
- Apex Triggers and Trigger Handler
- Batch Apex
- Scheduled Apex
- SOQL
- Salesforce Email Automation

6. Salesforce Application

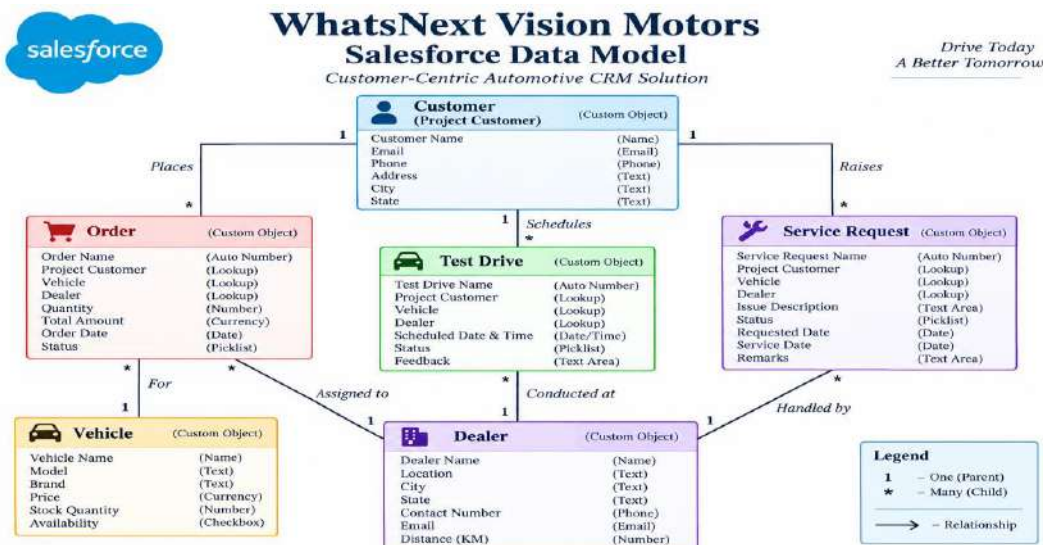
A Lightning application named WhatsNext Vision Motors was created with navigation for Home, Vehicles, Dealers, Customers, Orders, Test Drives, and Service Requests.



7. Data Model

The project contains the following major custom objects:

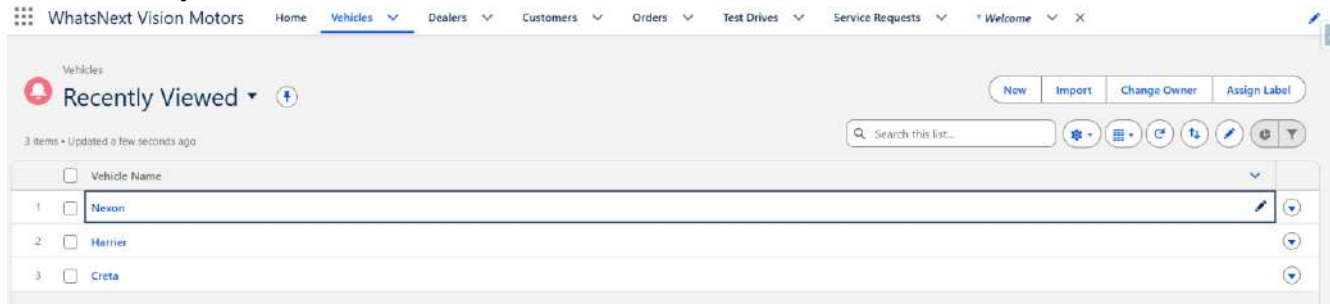
- Vehicle
- Dealer
- Customer
- Order
- Test Drive
- Service Request



8. Object and Field Details

Vehicle

- Model
- Brand
- Price
- Stock Quantity
- Availability



Dealer

- Dealer Name
- City
- State
- Street
- Latitude
- Longitude

WhatsNext Vision Motors Home Vehicles **Dealers** Customers Orders Test Drives Service Requests Welcome

Dealers

Recently Viewed

3 items • Updated a few seconds ago

New Import Change Owner Assign Label

Dealer Name

1	☐ Hyderabad Central	
2	☐ Secunderabad Motors	
3	☐ Gachibowli Motors	

WhatsNext Vision Motors Home Vehicles **Dealers** Customers Orders Test Drives Service Requests Welcome

Dealer

Hyderabad Central

Related Details

Dealer Name	Hyderabad Central	Owner	Jahnavi Pasupula
City	Hyderabad		
State	Telangana		
Street			
Latitude	17.385000		
Longitude	78.486700		
Dealer			
Created By	Jahnavi Pasupula : 9/4/2026, 8:48 AM	Last Modified By	Jahnavi Pasupula : 9/4/2026, 8:48 AM

Customer

- Customer Name
- Street
- City
- State
- Postal Code
- Latitude
- Longitude
- Phone
- Email

WhatsNext Vision Motors Home Vehicles Dealers **Customers** Orders Test Drives Service Requests Welcome

Customers

All

5 items • Sorted by Customer Name • Updated a few seconds ago

New Import Change Owner Printable View Assign Label

Customer Name

1	☐ Arijan Kumar	
2	☐ Aryan	
3	☐ Priya Reddy	
4	☐ Rahul Sharma	
5	☐ Vaishnavi	

- Project Customer
- Vehicle
- Dealer
- Scheduled Date & Time
- Status
- Reminder Sent

Test Drives

Recently Viewed

4 Items • Updated a few seconds ago

	Test Drive Number
1	☐ TD-0009
2	☐ TD-0008
3	☐ TD-0010
4	☐ TD-0007

Service Request

- Service Request Number
- Project Customer
- Vehicle
- Dealer
- Status
- Request Details

Service Requests

Recently Viewed

3 Items • Updated a few seconds ago

	Service Request Number
1	☐ SR-0003
2	☐ SR-0002
3	☐ SR-0001

Service Request

SR-0003

New Contact Edit

Related Details

Service Request Number
SR-0003

Project Customer
Rahul Sharma

Vehicle
Nexus

Request Type
Service

Description

Priority
Medium

Status
In Progress

Created By
Jahnavi Pasupula 9/12/2026, 4:30 AM

Owner
Jahnavi Pasupula

Last Modified By
Jahnavi Pasupula 9/12/2026, 4:30 AM

Activity

New Contact Edit

Filters: All time • All activity

Refresh Ex

Upcoming & Overdue

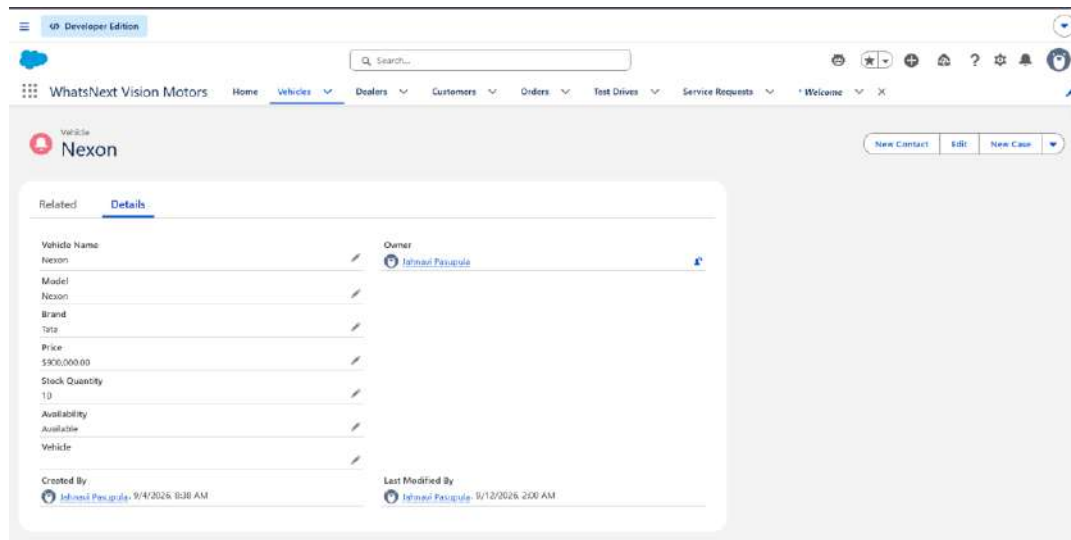
No activities to show.
Get started by sending an email, scheduling a task.

To change what's shown, try changing your filters.

Show All Activities

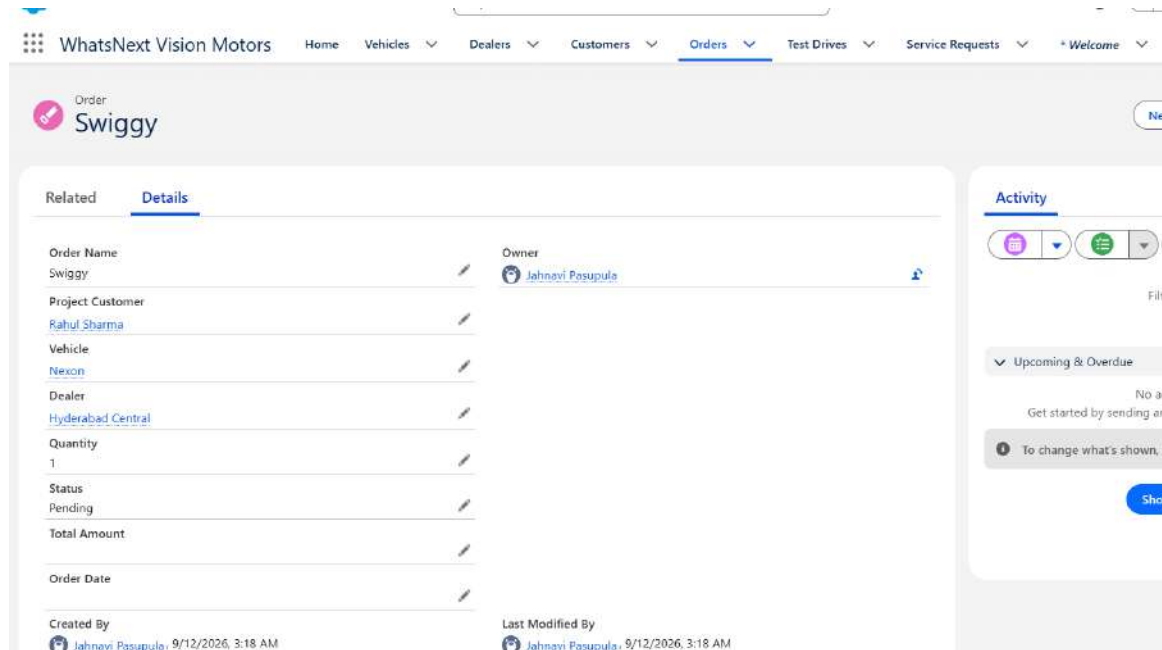
9. Vehicle Inventory Management

Vehicle records contain model, brand, price, stock quantity, and availability. The project uses stock quantity to determine whether a vehicle is Available or Out of Stock.



10. Order Management

Orders connect the project customer, vehicle, and dealer. Each order also stores quantity, order date, status, and total amount.



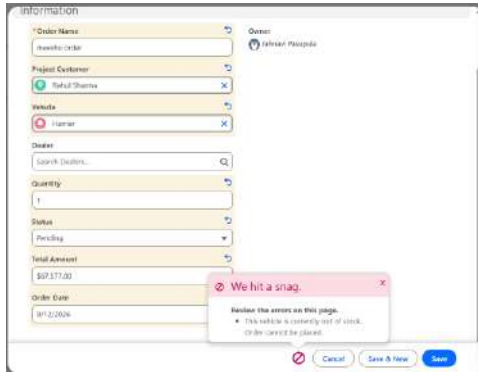
11. Stock Validation using Apex Trigger

An Apex trigger calls OrderTriggerHandler before an order is inserted. The handler checks the selected vehicle's Stock Quantity. If stock is zero or unavailable, the order is blocked with a validation message.

Expected validation message:

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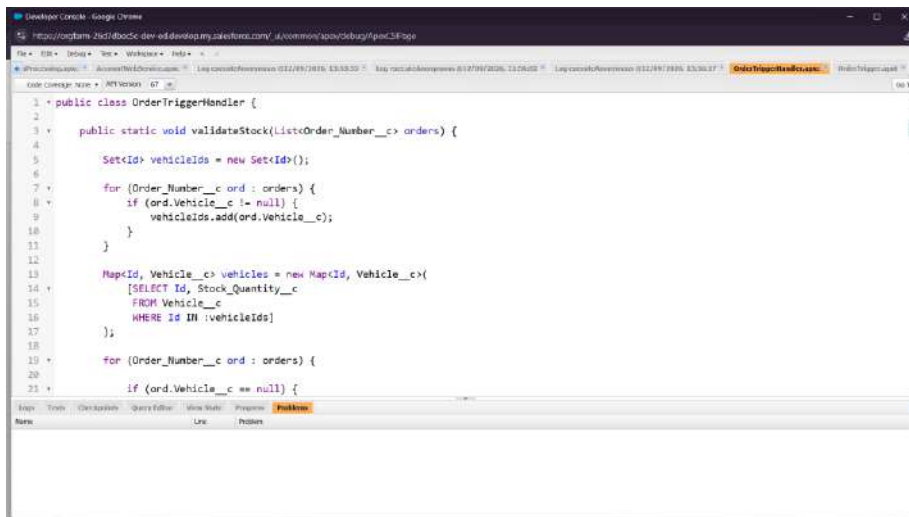
"This vehicle is currently out of stock. Order cannot be placed."



The screenshot shows a Salesforce 'New Order' form. The form includes fields for Order Name, Project Customer, Vehicle, Dealer, Quantity, Status, Total Amount, and Order Date. A red error message box is displayed over the form, stating: 'We hit a snag. Review the errors on this page. This vehicle is currently out of stock. Order cannot be placed.' The error message has a close button and a 'See' button.

Code Coverage: None API Version: 67

```
trigger OrderTrigger on Order_Number__c (before insert) {  
    OrderTriggerHandler.validateStock(trigger.new);  
    OrderTriggerHandler.assignDealer(trigger.new);  
}
```



12. Automatic Nearest Dealer Assignment

The OrderTriggerHandler retrieves the customer's latitude and longitude and compares them with dealer location coordinates. The dealer with the shortest calculated coordinate distance is assigned automatically to the order.

Dealer

[Hyderabad Central](#)

Quantity

Related

Details

Dealer Name	Hyderabad Central	Owner	Jahnavi Pasupula
City	Hyderabad		
State	Telangana		
Street			
Latitude	17.385000		
Longitude	78.486700		
Dealer			
Created By	Jahnavi Pasupula, 9/4/2026, 8:49 AM	Last Modified By	Jahnavi Pasupula, 9/4/2026, 8:48 AM

13. Test Drive Management

Customers can be associated with a vehicle, dealer, scheduled date and time, and test drive status. Scheduled test drives are processed by a record-triggered Flow.

WhatsNext Vision Motors

Home

Vehicles

Dealers

Customers

Orders

Test Drives

Service Requests

Welcome

Test Drive

TD-0009

New Contact Edit New Case

Related

Details

Text Drive Number

TD-0009

Owner

Jahnavi Pasupula

Project Customer

Rahul Sharma

Vehicle

Nexon

Dealer

Hyderabad Central

Scheduled Date & Time

9/12/2026, 4:00 PM

Status

Scheduled

Reminder Sent

Created By

Jahnavi Pasupula, 9/12/2026, 2:54 AM

Last Modified By

Jahnavi Pasupula, 9/12/2026, 2:54 AM

Activity

Filters: All time All activities All types

Refresh Expand All View All

Upcoming & Overdue

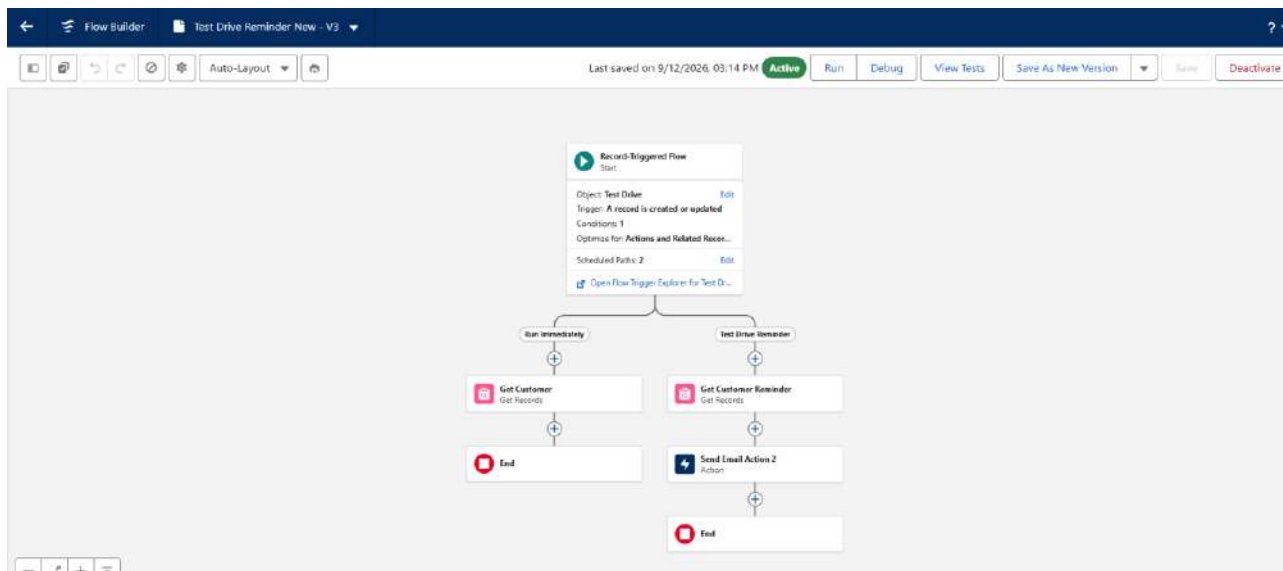
No activities to show. Get started by sending an email, scheduling a task, and more.

To change what's shown, try changing your filters.

Show All Activities

14. Test Drive Reminder Automation

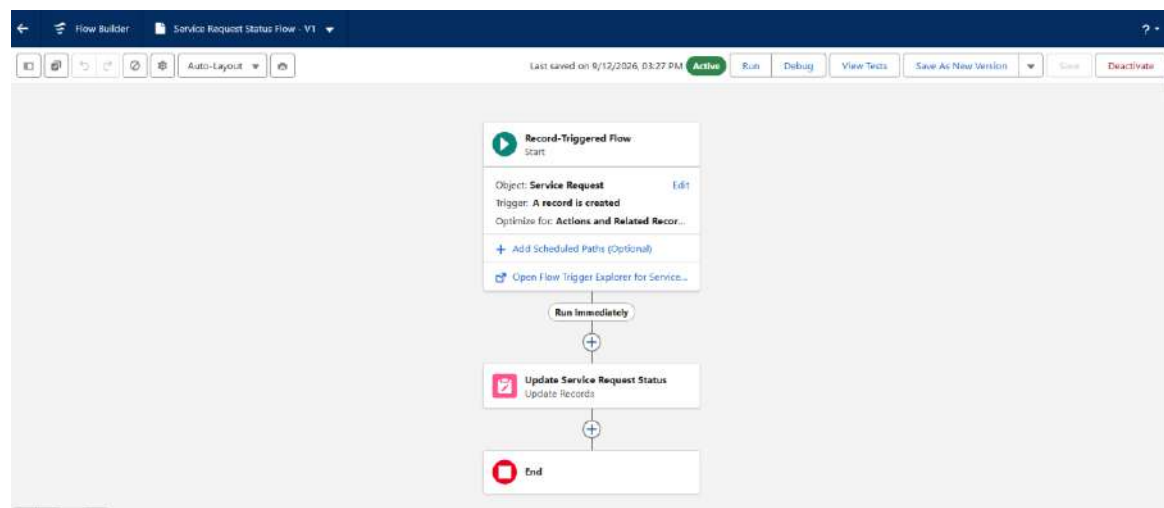
The Test Drive Reminder New Flow uses a scheduled path based on Scheduled Date & Time. The reminder is configured to run one hour before the scheduled test drive and send an email to the customer's email address.



15. Service Request Automation

A record-triggered Flow named Service Request Status Flow automatically updates the status of a newly created service request to In Progress.

Priority	Medium	
Status	In Progress	
Created By	Jahnavi Pasupula, 9/12/2026, 4:30 AM	Last Modified By: Jal



16. Batch Apex – Vehicle Stock Update

VehicleStockBatch processes vehicle records in batches and updates the Availability field. Vehicles with stock greater than zero are marked Available; vehicles with zero stock are marked Out of Stock.

```
global class VehicleStockBatch implements Database.Batchable<SObject> {

    global Database.QueryLocator start(Database.BatchableContext bc) {

        return Database.getQueryLocator(

            'SELECT Id, Stock_Quantity__c, Availability__c FROM Vehicle__c'

        );

    }

    global void execute(Database.BatchableContext bc, List<Vehicle__c> vehicles) {

        for (Vehicle__c vehicle : vehicles) {

            if (vehicle.Stock_Quantity__c != null &&

                vehicle.Stock_Quantity__c > 0) {

                vehicle.Availability__c = 'Available';

            } else {

                vehicle.Availability__c = 'Out of Stock';

            }

        }

        update vehicles;

    }

    global void finish(Database.BatchableContext bc) {

    }

}
```

17. Scheduled Apex – Daily Vehicle Stock Update

VehicleStockScheduler schedules the VehicleStockBatch to execute automatically. The configured job runs daily to keep vehicle availability synchronized with stock quantity.

VehicleStockScheduler Code:

```
global class VehicleStockBatch implements Database.Batchable<SObject> {

    global Database.QueryLocator start(Database.BatchableContext bc) {

        WhatsNext Vision Motors | Salesforce CRM Project | P. Jahnavi
```

```
global void execute(Database.BatchableContext bc, List<Vehicle__c> vehicles) {

    for (Vehicle__c vehicle : vehicles) {

        if (vehicle.Stock_Quantity__c != null &&
            vehicle.Stock_Quantity__c > 0) {

            vehicle.Availability__c = 'Available';

        } else {

            vehicle.Availability__c = 'Out of Stock';

        }

    }

    update vehicles;

}

global void finish(Database.BatchableContext bc) {

}
```

Q scheduled

- ✓ Environments
 - ✓ Jobs

Scheduled Jobs

didn't find what you're looking for? Try using Global Search.

18. Batch Apex – Order Status Processing

ORDER STATUS BATCH CODE:

```

global class OrderStatusBatch implements Database.Batchable<SObject> {

    global Database.QueryLocator start(Database.BatchableContext bc) {

        return Database.getQueryLocator(

            'SELECT Id, Vehicle__c, Status__c FROM Order_Number__c ' +

            'WHERE Status__c = \'Pending\'

        );

    }

    global void execute(Database.BatchableContext bc, List<Order_Number__c> orders) {

        Set<Id> vehicleIds = new Set<Id>();

        for (Order_Number__c ord : orders) {

            if (ord.Vehicle__c != null) {

                vehicleIds.add(ord.Vehicle__c);

            }

        }

        Map<Id, Vehicle__c> vehicles = new Map<Id, Vehicle__c>(

            [SELECT Id, Stock_Quantity__c

            FROM Vehicle__c

            WHERE Id IN :vehicleIds]

        );

        for (Order_Number__c ord : orders) {

            Vehicle__c vehicle = vehicles.get(ord.Vehicle__c);

            if (vehicle != null && vehicle.Stock_Quantity__c > 0) {

                ord.Status__c = 'Confirmed';

            } else {

                ord.Status__c = 'Pending';

            }

        }

    }

}

```

```

        update orders;
    }

    global void finish(Database.BatchableContext bc) {
    }
}

```

19. Scheduled Stock and Order Notifications

StockOrderNotificationScheduler checks for vehicles with zero stock and orders that are still Pending. If either condition exists, the scheduler sends an email notification to the current Salesforce user's email address.

STOCK/ORDER NOTIFICATION SCHEDULER CODE:

```

global class StockOrderNotificationScheduler implements Schedulable {

    global void execute(SchedulableContext sc) {

        List<Vehicle__c> outOfStockVehicles = [

            SELECT Name, Model__c, Stock_Quantity__c

            FROM Vehicle__c

            WHERE Stock_Quantity__c = 0

        ];

        List<Order_Number__c> pendingOrders = [

            SELECT Name, Status__c

            FROM Order_Number__c

            WHERE Status__c = 'Pending'

        ];

        if (outOfStockVehicles.isEmpty() && pendingOrders.isEmpty()) {

            return;

        }

        User currentUser = [

            SELECT Email

            FROM User

```

```

        WHERE Id = :UserInfo.getUserId()

    ];

    String body = 'WhatsNext Vision Motors Notification\n\n';

    if (!outOfStockVehicles.isEmpty()) {

        body += 'STOCK REPLENISHMENT REQUIRED:\n';

        for (Vehicle__c vehicle : outOfStockVehicles) {

            body += '- ' + vehicle.Model__c +

                ' | Stock: ' + vehicle.Stock_Quantity__c + '\n';

        }

        body += '\n';

    }

    if (!pendingOrders.isEmpty()) {

        body += 'PENDING ORDER PROCESSING:\n';

        for (Order_Number__c orderRecord : pendingOrders) {

            body += '- Order: ' + orderRecord.Name +

                ' | Status: ' + orderRecord.Status__c + '\n';

        }

    }

    Messaging.SingleEmailMessage email =

        new Messaging.SingleEmailMessage();

    email.setToAddresses(new String[] { currentUser.Email });

    email.setSubject('WhatsNext Vision Motors - Stock & Order Notification');

    email.setPlainTextBody(body);

    Messaging.sendEmail(

        new Messaging.SingleEmailMessage[] { email }

    );

}

}

```

20. Testing and Results

The following test cases can be documented.

Out-of-stock order: Input – Attempted to order Harrier with stock quantity 0. Result – Order blocked with out-of-stock error.

In-stock order: Input – Created an order for an available vehicle. Result – Order created successfully.

Nearest dealer assignment: Input – Created an order with a project customer and vehicle. Result – Dealer was automatically assigned.

Vehicle stock batch: Input – Executed VehicleStockBatch. Result – Availability updated according to stock quantity.

Pending order processing: Input – Executed OrderStatusBatch. Result – Orders with available stock changed to Confirmed.

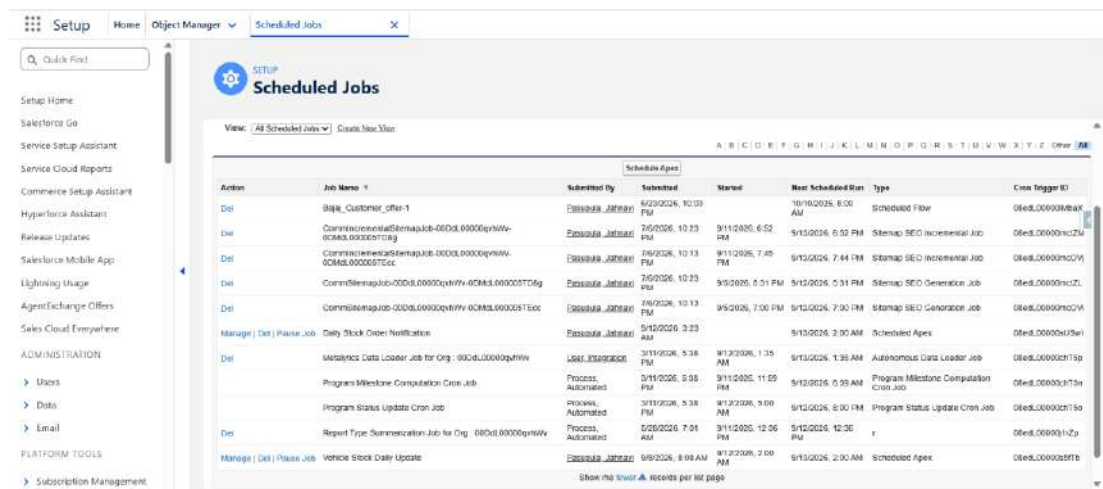
Test drive reminder: Input – Scheduled a test drive. Result – Reminder Flow configured to send email one hour before the scheduled time.

Service request automation: Input – Created a service request. Result – Status changed automatically to In Progress.

21. Scheduled Jobs

Configured scheduled processes include:

- Vehicle Stock Daily Update
- Daily Stock Order Notification

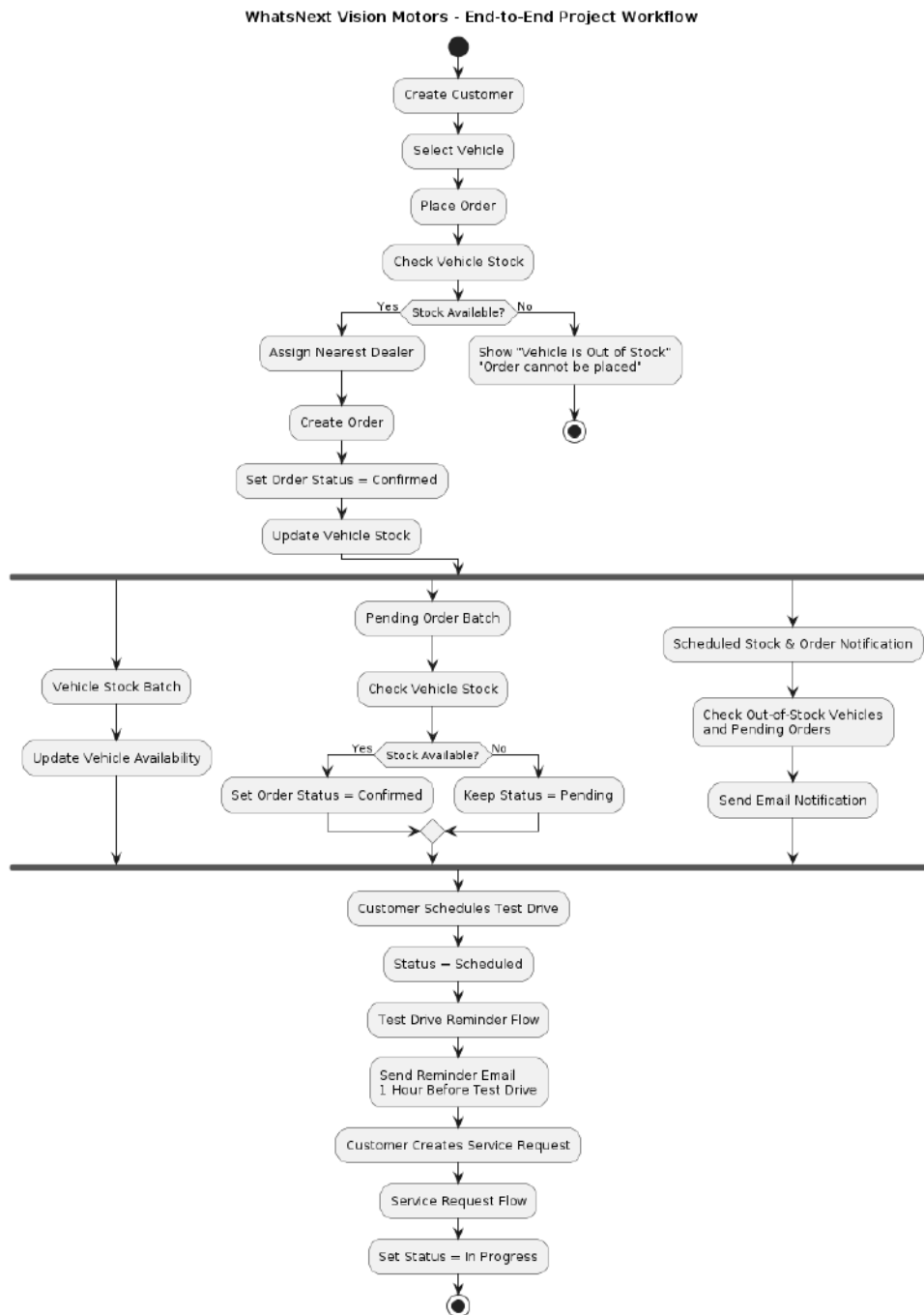


Action	Job Name	Submitted By	Submitted	Started	Next Scheduled Run	Type	Cron Trigger ID
Del	Bqle_Customer_offer-1	Process, Automated	5/23/2026, 10:03 PM	5/11/2026, 4:52 PM	5/13/2026, 5:00 AM	Scheduled Flow	06e8.000000bax
Del	CommIncrementalStemapJob-000d.0000qynWw-0CMd.000000T0g	Process, Automated	7/5/2026, 10:23 PM	5/11/2026, 7:45 PM	5/13/2026, 5:52 PM	Stemap SEO Incremental Job	06e8.000000iczu
Del	CommIncrementalStemapJob-000d.0000qynWw-0CMd.000000T0g	Process, Automated	7/5/2026, 10:13 PM	5/11/2026, 7:45 PM	5/13/2026, 7:44 PM	Stemap SEO Incremental Job	06e8.000000icv
Del	CommStemapJob-000d.0000qynWw-0CMd.000000T0g	Process, Automated	7/5/2026, 10:23 PM	5/13/2026, 5:51 PM	5/13/2026, 5:51 PM	Stemap SEO Generation Job	06e8.000000iczi
Del	CommStemapJob-000d.0000qynWw-0CMd.000000T0g	Process, Automated	7/5/2026, 10:13 PM	5/13/2026, 7:30 PM	5/13/2026, 7:30 PM	Stemap SEO Generation Job	06e8.000000icv
Manage Del Pause Job	Daily Stock Order Notification	Process, Automated	5/12/2026, 3:23 AM		5/13/2026, 2:00 AM	Scheduled Apex	06e8.000000d5w
Del	Metastore Data Loader Job for Org : 000d.0000qynWw	Process, Automated	3/11/2026, 5:38 PM	5/12/2026, 1:35 AM	5/13/2026, 1:35 AM	Autonomous Data Loader Job	06e8.000000ch5p
	Program Milestone Computation Cron Job	Process, Automated	3/11/2026, 5:05 PM	5/11/2026, 11:05 PM	5/12/2026, 5:05 AM	Program Milestone Computation Cron Job	06e8.000000j75n
	Program Status Update Cron Job	Process, Automated	3/11/2026, 5:38 PM	5/12/2026, 5:09 AM	5/12/2026, 5:00 PM	Program Status Update Cron Job	06e8.000000n15o
Del	Report Type Summation Job for Org : 000d.0000qynWw	Process, Automated	5/25/2026, 7:01 AM	5/11/2026, 12:06 PM	5/12/2026, 12:06 PM	r	06e8.000000i2p
Manage Del Pause Job	Vehicle Stock Daily Update	Process, Automated	5/9/2026, 8:09 AM	5/12/2026, 2:00 AM	5/13/2026, 2:00 AM	Scheduled Apex	06e8.000000bfb

22. Project Workflow

Customer information is maintained in Salesforce. When a customer places an order, the system validates vehicle stock and automatically assigns a nearby dealer. Vehicle inventory is periodically checked using Batch

Apex and Scheduled Apex. Pending orders are processed when stock becomes available. Test drives receive automated reminder emails, while service requests are automatically moved to In Progress.



23. Learning Outcomes

- Understanding Salesforce CRM and Lightning applications.
- Creating custom objects, fields, tabs, and relationships.
- Designing Salesforce data models.
- Building record-triggered Flows and scheduled paths.

- Developing Apex classes and triggers.
- Implementing trigger-handler architecture.
- Using SOQL for querying related Salesforce records.
- Developing Batch Apex for bulk processing.
- Scheduling Apex jobs for periodic automation.
- Implementing automated email notifications.
- Testing Salesforce functionality using real project records.

25. Future Enhancements

- Use a more advanced geographic distance calculation or Salesforce geolocation fields for dealer matching.
- Create dashboards for sales, inventory, test drives, and service requests.
- Add role-based access and sharing rules for different dealership users.
- Integrate external vehicle or inventory systems.
- Add customer self-service capabilities through Experience Cloud.
- Add richer reporting and analytics for dealership performance.

26. Demo Link

The project demonstration is provided through a hosted demo video. Paste the final Google Drive or other accessible demo URL below.

<G:\My Drive\Salesforce SkillWallet Project.mp4>

27. GitHub Repository

The GitHub repository contains the project documentation and selected implementation files such as Apex classes, trigger code, and supporting project information.

28. Salesforce Trailhead Profile

Salesforce Trailhead Profile:

<https://www.salesforce.com/trailblazer/dyjyzvcoc58m0ilgdf>

29. Conclusion

WhatsNext Vision Motors demonstrates how Salesforce can be used to build a structured CRM solution for vehicle dealership operations. The project combines Salesforce data modeling, Lightning applications, Flow automation, Apex triggers, Batch Apex, Scheduled Apex, and email notifications to reduce manual work and improve order, inventory, test-drive, and service-request management.

Thankyou.